

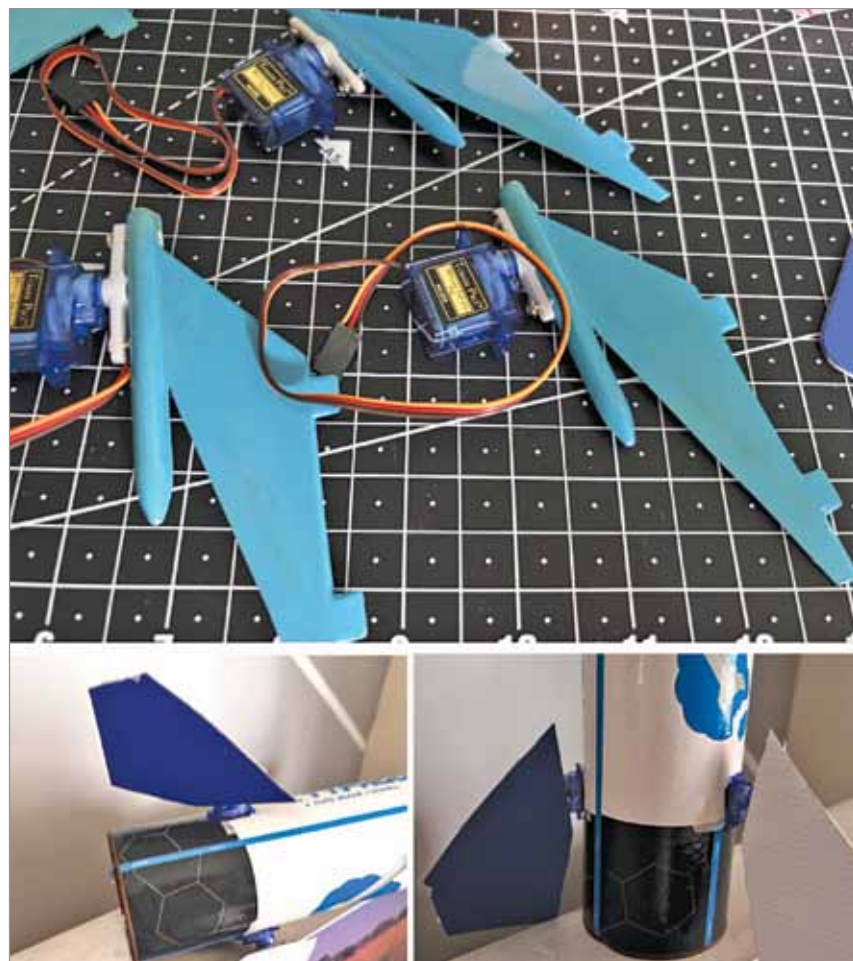


Fig. 6: Model rocket with servo-controlled fin assembly



Fig. 7: Rocket fins moving for stabilisation

section of the rocket body.

Attach a fin securely to the servo horn (servo shaft). Ensure that the fin is firmly fixed and can rotate smoothly without obstruction. Once the fin is attached, mount the servo motor inside the rocket body so that the fin protrudes through the fin slot (see Fig. 6) and can move freely. Repeat the same procedure for the remaining fins, where applicable.

Although some rocket designs

use multiple independently controlled fins for improved flight control and stabilisation, this system uses only the bottom fins with servo actuation to keep the design simple, economical, and easy to reproduce. More advanced designs may employ additional servo-controlled fins to achieve greater manoeuvrability and improved flight stability.

EFY note. Fin size, shape, and placement depend on the rocket design. The fin assembly shown here is intended for demonstration and testing only. Optimise the fin design and its position based on the simulation results to achieve the best flight performance.

Testing

The system is now ready for testing. Begin with software and telemetry testing.

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

Power on the rocket to enable its Wi-Fi access point (AP). Connect a laptop or smartphone to the rocket's Wi-Fi network, then open the web dashboard to monitor telemetry data in real time. Fig. 7 shows the rocket fins moving for stabilisation.

The web dashboard displays the stabilisation system in operation. As the rocket body is tilted or moved, the control fins automatically adjust their positions using the PID algorithm to counter the movement and stabilise the rocket.

The dashboard displays real-time information, including altitude, IMU data, barometric pressure, temperature, PID output, fin positions, and other sensor readings.

Precautions.

1. Conduct all tests only in a vacant ground with a minimum clear space of 150 metres in all directions.
2. Ensure the rocket is properly assembled and that all safety procedures are strictly followed under the guidance of an experienced engineer.
3. To begin with, test only the rocket's software, electronics, telemetry, and stabilisation systems. Do not perform a full flight test.

4. Before conducting any test, carefully review the NASA model rocket safety guide:

<https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/model-rocket-safety/>

5. Failure to follow these instructions may result in serious injury, property damage, or legal consequences. **EFY**

Ashwini Kumar Sinha, an IoT and AI enthusiast, is Tech Journalist at EFY.